E-waste It Wisely at the Sub-National Level: A District-Level Assessment of Generation, Collection, and Recycling Performance in Sikkim, India

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Abstract—

E-waste is among the fastest-growing domestic waste streams globally, and sub-national jurisdictions in developing regions face acute challenges in generation monitoring, collection, and formal recycling. This paper presents a district-level assessment of e-waste management performance in the state of Sikkim, India, for financial year 2026, drawing on data reported by the Sikkim State Pollution Control Board under the E-Waste (Management) Rules, 2022. Across the six districts of Sikkim, an estimated 310.5 tonnes of e-waste was generated, of which only 18.2 tonnes (5.9%) was formally collected, indicating a substantial capture gap consistent with patterns observed in other developing regions. Extended Producer Responsibility (EPR) achievement stood at 83.9% of target, with 12 of 20 registered producers fully compliant. Three authorised recycling facilities, with an installed capacity of 1,100 tonnes per year, processed 690 tonnes with material recovery rates of 77-81%. Only 42% of identified informal-sector e-waste workers have been registered or trained, and awareness campaigns have not yet reached two of the six districts. The findings suggest that, in common with other developing regions, the principal constraint on effective e-waste management in Sikkim is not legislative absence but weak collection infrastructure, uneven district coverage, and limited formalisation of the informal sector. Recommendations are offered for expanding collection infrastructure, extending recycler and awareness coverage to underserved districts, and formalising informal-sector labour.

Keywords— *E-waste; WEEE; Extended Producer Responsibility; Sikkim; solid waste management; informal recycling; India.*

I. INTRODUCTION

Electrical and electronic equipment (EEE) waste, or e-waste, is widely recognised as the world's fastest-growing domestic waste stream, driven by rising consumption, rapid product obsolescence, and expanding access to electronics [1]. Globally, e-waste volumes are estimated to have more than doubled over the past decade, and a large share of this waste remains undocumented at end-of-life, particularly in developing and transitioning economies where collection and recycling infrastructure lags behind generation [1], [4]. Case evidence from Africa illustrates how gaps in monitoring, informal-sector dominance, and uneven legislative coverage can allow e-waste to accumulate in poorly managed dumpsites, with attendant risks to environmental and human health [1]. Similar structural pressures, on a smaller scale, are visible in sub-national jurisdictions of large developing economies such as India, where state-level Pollution Control Boards are increasingly tasked with reporting generation, collection, and recycling outcomes in a standardised, auditable form.

India regulates e-waste under the E-Waste (Management) Rules, 2022, which assign Extended Producer Responsibility (EPR) targets to producers and require state-level Pollution Control Boards to monitor generation, collection, and authorised recycling within their jurisdiction. Sikkim, a hill state in India's Northeast with a small, dispersed population and mountainous terrain, presents a distinctive case: its six administrative districts vary sharply in population density, urbanisation, and accessibility, factors that plausibly shape both e-waste generation and the feasibility of formal collection. Sikkim reports its performance through a district-level dashboard maintained by the Sikkim State Pollution Control Board (SPCB).

This paper synthesises that dashboard, covering the period ending 30 July 2026, into a structured assessment of generation, collection, compliance, and recycling infrastructure across Sikkim's six districts: Gangtok, Gyalshing, Mangan, Namchi, Pakyong, and Soreng. The objective is threefold: (i) to quantify how e-waste generation, collection, and recycling capacity are distributed across Sikkim's districts; (ii) to assess whether reported Extended Producer Responsibility performance is consistent with on-the-ground collection outcomes; and (iii) to identify where Sikkim's experience mirrors, or diverges from, patterns documented in the wider e-waste literature, drawing in particular on comparative evidence from the African context [1]. The analysis is intended to support evidence-based prioritisation of collection infrastructure, recycler licensing, and outreach investment across Sikkim's districts.

A. E-waste: Definitions and Categories

Consistent with international usage, e-waste in this paper refers to discarded electrical and electronic equipment (EEE) and its parts, encompassing items intentionally discarded without intent of reuse (waste EEE, or WEEE) as well as used EEE that remains functional or repairable but has reached the end of its first use cycle [1]. The Sikkim SPCB dashboard classifies generation into six categories: large and small equipment (e.g., refrigerators, washing machines, and small household appliances), information technology and telecommunications equipment, consumer electronics, batteries, lighting equipment, and an unclassified residual "other" category. This categorisation broadly follows the six-category schema used in international e-waste monitoring [4]

and enables comparison with generation-mix patterns reported at national and global scale.

II. REGULATORY AND COMPARATIVE CONTEXT

India's E-Waste (Management) Rules, 2022 replace earlier 2016 rules and place binding annual collection and recycling targets on producers, expressed as a percentage of the quantity of EEE placed on the market a fixed number of years earlier. Producers discharge this Extended Producer Responsibility (EPR) obligation by purchasing EPR certificates from registered recyclers, dismantlers, or refurbishers, and are required to file returns through a centralised online portal. State Pollution Control Boards, including the Sikkim SPCB, are responsible for registering collection centres and recyclers operating within their jurisdiction, monitoring producer compliance, and reporting consolidated state-level performance under Schedule I of the Rules.

This regulatory design is broadly analogous, in intent if not in institutional form, to the combination of international and regional instruments that govern transboundary e-waste movement elsewhere, such as the Basel Convention and its regional counterpart the Bamako Convention in Africa [1]. Both frameworks share a common objective: shifting the cost of end-of-life management back onto producers or importers, and creating a documented chain of custody from generation through to formal recycling. Evidence from Africa, however, shows that legislative coverage alone is an incomplete predictor of outcomes; enforcement capacity, monitoring infrastructure, and the scale of the informal sector often matter more than the mere existence of a legal framework [1]. This distinction motivates the present paper's focus on operational indicators, capture rates, recycler utilisation, informal-worker registration, rather than on regulatory design alone.

At a global scale, e-waste generation reached over 53 million tonnes in 2019, of which only 17.4% was documented as formally collected and recycled, down from 20% in 2016 despite a 21% growth in volumes over the same period [4]. Africa's e-waste generation for the same year was estimated at 2.9 million tonnes domestically produced, with substantial additional volumes arriving through transboundary imports, and an average per-capita generation of 2.5 kg per year, well below the Americas (13.3 kg) and Europe (16.2 kg) [1], [4]. These figures provide a useful, if imperfect, benchmark against which Sikkim's own capture rate and per-capita generation can be read, and are referenced in the Discussion below.

III. DATA AND METHODOLOGY

The analysis is based entirely on secondary data extracted from the Sikkim SPCB "E-Waste Management Dashboard" (FY 2026), reported under Schedule I of the E-Waste (Management) Rules, 2022. The dashboard compiles figures from the District Generation Inventory, the Collection Log and Compliance Tracker, the Recycling Infrastructure register, the Informal Sector Hotspot survey, the Environmental Health Monitoring programme, and the Awareness Programme records, covering all six districts of Sikkim. Data reflect the period as of 24-30 July 2026 and are presented at both state and district level. No independent field verification was undertaken for this paper; the analysis is limited to interpretation of the reported figures.

IV. RESULTS

A. Generation and Collection

Total e-waste generation across Sikkim for FY 2026 is reported at 310.5 tonnes (310,500 kg). Generation is heavily concentrated in Gangtok district, which alone accounts for 120 tonnes, or 39% of the state total, reflecting the concentration of urban population, offices, and institutions in the state capital. The remaining districts each generate considerably less: Namchi and Gyalshing report 52.5 tonnes each, Pakyong 31.5 tonnes, Mangan 28.0 tonnes, and Soreng 26.0 tonnes (Fig. 1, Table I).

Against this generation, only 18.2 tonnes (5.9%) was formally collected statewide. Collection lags generation by roughly an order of magnitude in every district without exception, indicating that the great majority of e-waste generated in Sikkim does not currently enter a formal collection channel. This pattern mirrors findings from other developing regions, where large discrepancies between waste produced and waste properly collected point to substantial undocumented flows into informal repair, storage, or disposal routes [1]. Notably, the district with the highest absolute collection volume, Gangtok (8.2 t), still collects only 6.8% of its own generation, showing that even Sikkim's best-resourced district falls far short of adequate capture; smaller districts such as Mangan (0.9 t collected against 28.0 t generated, a 3.2% capture rate) fare considerably worse.

B. Generation Mix by Category

By category, large and small equipment together constitute the largest share of generated e-waste (101.3 t, 33%), followed by IT and telecommunications equipment (83.0 t, 27%) and consumer electronics (72.5 t, 23%). Batteries (28.1 t, 9%) and lighting (13.8 t, 4%) form smaller shares, with an unclassified "other" category accounting for the remainder (Fig. 1). The dominance of large and small equipment is significant because these categories are typically bulkier, more likely to contain hazardous components such as refrigerants and printed circuit boards, and less likely to be captured by informal kerbside or retailer take-back arrangements than smaller IT items.

Batteries warrant particular attention despite their comparatively modest 9% share by weight: lithium-ion and

lead-acid batteries pose fire and chemical-leaching risks disproportionate to their mass if landfilled or informally dismantled, and typically require dedicated collection streams separate from bulk e-waste.

C. Extended Producer Responsibility (EPR) Achievement

Statewide EPR achievement stood at 83.9% (12,500 kg against a 14,900 kg target). Performance varies by district: Namchi and Mangan report achievement closest to target, while other districts trail more substantially. Aggregated EPR performance therefore masks considerable district-level unevenness, and the gap between statewide EPR achievement (83.9%) and the far lower formal collection capture rate (5.9%) suggests that EPR reporting may draw on channels distinct from the district collection-centre network captured elsewhere in the dashboard, for example, EPR certificates purchased against recycling performed on e-waste collected outside Sikkim, or against processing by the state's three authorised recyclers independent of the district collection-centre count.

D. Producer Compliance

Of 20 registered producers, 12 (60%) are fully compliant, 6 (30%) remain partially compliant pending inspection or documentation, and 2 (10%), Lenovo and OPPO, are non-compliant, with shortfalls in collection targets and penalty action initiated against both. The dashboard identifies the non-compliant producers by name, indicating that compliance monitoring at the producer level is granular and individually actionable, which is a comparatively mature feature relative to jurisdictions where producer-level accountability is difficult to establish. The six partially-compliant producers awaiting inspection or documentation (Dell, ASUS, Xiaomi, Realme, Haier, and Havells) represent a further 30% of the registered base whose ultimate compliance status remains to be confirmed.

E. Recycling Infrastructure

Sikkim has three authorised recycling facilities with a combined installed capacity of 1,100 tonnes per year (Table II): Sikkim Eco Dismantlers (Gangtok, 500 t/yr capacity, 320 t processed, 81% recovery), Eastern Recyclers (Namchi, 350 t/yr, 210 t processed, 79% recovery), and Himalayan E-Waste Solutions (Pakyong, 250 t/yr, 160 t processed, 77% recovery). Combined utilisation across the three facilities is approximately 63%, and material recovery rates of 77-81% are consistent across facilities, suggesting that recycling quality, where recyclers are present, is broadly satisfactory. However, all three authorised facilities are located in only three of the six districts (Gangtok, Namchi, Pakyong); Gyalshing, Mangan, and Soreng currently have no authorised recycling capacity of their own. Combined installed capacity (1,100 t/yr) substantially exceeds current total generation (310.5 t/yr), implying that the binding constraint on Sikkim's e-waste system is upstream collection rather than downstream processing capacity: existing recyclers are running well below capacity (63% average utilisation) principally because too little material is being formally collected and routed to them.

F. Informal Sector Formalisation

The dashboard identifies 97 informal e-waste workers across three hotspot clusters (MG Marg Scrap Cluster, Namchi Scrap Market, Pakyong Repair Cluster), of whom only 41 (42%) are registered or trained. Registration shares are broadly similar across the three hotspots (40-47%), indicating that formalisation gaps are distributed relatively evenly rather than concentrated in a single site. This finding is consistent with the broader e-waste literature, in which informal collection, repair, and dismantling activity persists alongside formal systems even where recycling infrastructure exists, because informal actors offer convenience, immediacy of payment, and geographic reach that formal collection centres often cannot match [1].

G. Environmental Health Monitoring

Environmental monitoring at the three identified informal hotspots found air and soil quality within limits at the MG Marg Scrap Cluster and Namchi Scrap Market respectively, while the Pakyong Repair Cluster showed minor water contamination (Table III). Health camps were conducted at two of the three sites, screening a combined 195 individuals; no health camp had yet been conducted at the Namchi Scrap Market at the time of reporting. The absence of major contamination findings to date is a comparatively favourable result relative to long-established, high-volume informal e-waste sites documented elsewhere, where heavy metals, dioxins, and polycyclic aromatic hydrocarbons have been measured at elevated levels in soil, dust, and downstream water bodies after years of unregulated processing [1]. This likely reflects the still-modest scale of Sikkim's informal e-waste activity relative to such sites, rather than an absence of underlying risk, and argues for sustained monitoring before, rather than after, contamination becomes established.

H. Awareness and Outreach

Three awareness campaigns have been conducted, reaching a combined 1,580 people, with door-to-door collection drives in Gangtok (850 participants) forming the largest single channel, followed by a college e-waste campaign in Namchi (430) and a village awareness camp in Soreng (300). Gyalshing and Mangan have not yet been covered by any awareness programme. Households remain the largest audience segment reached to date, while institutional channels such as colleges have reached a comparatively smaller, though potentially high-multiplier, audience.

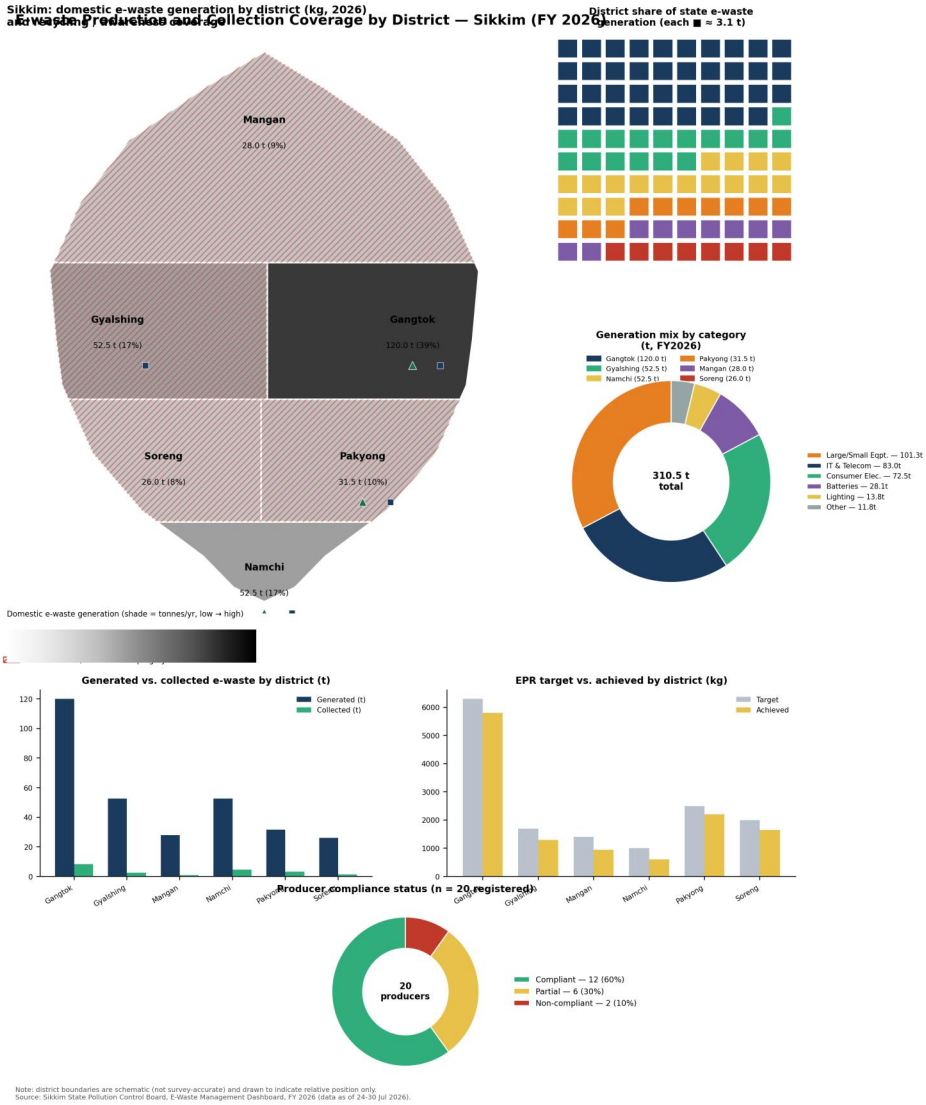


Fig. 1. District-level e-waste generation, collection, recycler and awareness coverage in Sikkim (FY 2026). Source: Sikkim SPCB E-Waste Management Dashboard.

V. DATA TABLES

TABLE I. DISTRICT-WISE GENERATION AND COLLECTION SUMMARY (FY 2026)

District	Generated (t)	Collected (t)	Capture rate	Recycler	Awareness prog.
Gangtok	120.0	8.2	6.8%	Yes	Yes
Gyalshing	52.5	2.6	5.0%	No	No
Namchi	52.5	4.7	9.0%	Yes	Yes
Pakyong	31.5	3.3	10.5%	Yes	No
Mangan	28.0	0.9	3.2%	No	No
Soreng	26.0	1.3	5.0%	No	Yes

Capture rate = collected / generated. Values derived from the Sikkim SPCB E-Waste Management Dashboard, FY 2026.

TABLE II. AUTHORISED RECYCLER DETAIL

Recycler	District	Capacity (t/yr)	Processed (t)	Recovery	Valid till
Sikkim Eco Dismantlers	Gangtok	500	320	81%	31-03-2028

Recycler	District	Capacity (t/yr)	Processed (t)	Recovery	Valid till
Eastern Recyclers	Namchi	350	210	79%	31-03-2029
Himalayan E-Waste Solutions	Pakyong	250	160	77%	31-03-2029

TABLE III. ENVIRONMENTAL HEALTH MONITORING SITES

Site	Type	Result	Health camp	Screened
MG Marg Scrap Cluster	Air quality	Within limit	Yes	120
Namchi Scrap Market	Soil	Within limit	No	0
Pakyong Repair Cluster	Water	Minor contamination	Yes	75

VI. DISCUSSION

Taken together, the results indicate that Sikkim's principal e-waste management challenge, in common with patterns documented elsewhere, is not an absence of regulatory or institutional structure but a substantial gap between waste generated and waste formally captured. A collection capture rate of 5.9% against generation, even allowing for informal repair and reuse absorbing some fraction of used equipment, implies that a large volume of e-waste in Sikkim is likely accumulating outside the formal system, whether in household storage, informal repair shops, or unmanaged disposal. This pattern is qualitatively consistent with the broader literature on e-waste in developing regions, where discrepancies between production and documented collection are attributed to under-resourced monitoring and a dominant informal sector [1]. It is worth noting that Sikkim's reported 5.9% capture rate is, on this specific metric, markedly lower than the 17.4% global average collection and recycling rate reported for 2019 [4], underlining that Sikkim's collection infrastructure has considerable room to mature even relative to a global benchmark that is itself widely regarded as inadequate.

District-level unevenness is a second recurring theme. Gangtok's concentration of both generation (39% of the state total) and recycling capacity mirrors the pattern in which major population and administrative centres act as both the largest source and the primary processing node for e-waste, while peripheral districts (Gyalshing, Mangan, Soreng) lack local recycling capacity and, in the case of Gyalshing and Mangan, any awareness outreach at all. This concentration risks leaving residents in less-served districts with fewer viable formal disposal options, which may in turn reinforce reliance on informal channels. The pattern is directly comparable to findings from Africa, where a small number of major ports and cities function as both generation hotspots and processing nodes, while surrounding areas remain comparatively underserved by formal infrastructure [1].

The informal sector plays a load-bearing role in Sikkim's e-waste economy, as it does in many developing regions [1], yet only 42% of identified workers are registered or trained. Given that informal actors are frequently the first point of contact for discarded electronics, their formalisation, through registration, training, and provision of protective equipment, represents a comparatively low-cost lever for improving both worker safety and the quality of downstream material recovery. Elsewhere, informal e-waste recycling has been documented to provide meaningful livelihoods at significant scale, for example, generating tens of thousands of jobs in some West African e-waste economies, even as its rudimentary processing methods create elevated health and environmental risk for workers and surrounding communities [1]; Sikkim's much smaller informal workforce (97 identified workers) suggests the state is at an early stage of this dynamic, where proactive formalisation may be considerably easier to achieve than in jurisdictions where informal e-waste processing has become deeply entrenched.

The observed mismatch between recycling capacity (1,100 t/yr installed, 63% utilised) and formal collection (18.2 t/yr, 5.9% of generation) is, in the state's favour, a more tractable problem than a capacity shortfall would be: it implies that targeted investment in collection logistics, rather than in additional processing infrastructure, is likely to yield the largest near-term improvement in Sikkim's formal e-waste capture.

A. Limitations

This assessment is subject to several limitations inherent to its reliance on a single secondary dashboard source. First, no independent verification of reported generation, collection, or compliance figures was possible; the analysis assumes the underlying District Generation Inventory and Collection Log figures are accurate as reported. Second, generation estimates for a small, mountainous state with limited producer presence are inherently harder to model than for larger, more data-rich jurisdictions, and the methodology underlying the District Generation Inventory is not detailed in the dashboard itself. Third, district boundaries used in Fig. 1 are schematic and drawn to indicate relative geographic position only; they are not survey-accurate administrative boundaries. Finally, the dashboard's reporting window (24-30 July 2026) represents a single snapshot rather than a full-year trend, so seasonal

variation in generation and collection cannot be assessed from this data alone.

VII. RECOMMENDATIONS

- Expand formal collection infrastructure beyond the current eight registered collection centres, prioritising bulky large/small equipment, which forms the single largest generation category (33%) and is least likely to be captured informally.
- Extend authorised recycling capacity and collection-centre coverage to Gyalshing, Mangan, and Soreng, the three districts currently without either, using Gangtok's collection-centre-plus-recycler model as a template adapted to lower-volume districts (e.g., shared or mobile collection points rather than dedicated fixed facilities).
- Scale registration and training of informal-sector workers, targeting the 58% of identified workers (56 of 97) who remain unregistered, with attention to all three identified hotspot clusters rather than a single site, and with basic personal protective equipment provided as part of registration.
- Run targeted awareness campaigns in Mangan and Gyalshing, the two districts not yet reached by any outreach programme, potentially adapting the door-to-door model that reached 850 people in Gangtok.
- Investigate and reconcile the divergence between the 83.9% statewide EPR achievement figure and the 5.9% formal collection capture rate, to clarify which channels are contributing to reported EPR compliance and whether district collection-centre data should be better integrated into EPR reporting.
- Continue and expand environmental health monitoring, including a health camp at the Namchi Scrap Market, where none has yet been conducted, and follow-up monitoring of the minor water contamination identified at the Pakyong Repair Cluster, before the scale of informal processing grows further.
- Prioritise investment in collection logistics over additional processing capacity in the near term, given that installed recycling capacity (1,100 t/yr) already substantially exceeds current total generation (310.5 t/yr) and utilisation across all three recyclers remains at approximately 63%.
- Publish district-level dashboards on a recurring (e.g., quarterly) basis rather than as a single snapshot, to allow trend analysis and to track the effect of the above interventions over time.

VIII. CONCLUSION

Sikkim's E-Waste Management Dashboard for FY 2026 shows a state with functioning regulatory and monitoring institutions, an EPR achievement rate above 80%, and recycling facilities that, where present, achieve solid material recovery rates. At the same time, only 5.9% of generated e-waste is formally collected, recycling and awareness coverage remain concentrated in three of six districts, and more than half of identified informal workers remain outside formal registration or training. Closing these gaps, through expanded district-level collection infrastructure, targeted outreach to underserved districts, and continued formalisation of informal-sector labour, represents the most direct route to improving e-waste outcomes in Sikkim, consistent with lessons drawn from e-waste management challenges observed elsewhere [1].

IX. ACKNOWLEDGMENT

[Add acknowledgments here — e.g., data source and any institutional support.]

X. REFERENCES

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